

Implementing Real-Time Monitoring in an Event Management System with InfluxDB and Grafana

This in-depth explanation of the event management system designed and developed by the authors highlights the web-based platform's various capabilities, including logging, event management, and real-time analytics.



The event management system (EMS) described here is a web-based platform designed to handle event creation, user participation, and real-time performance monitoring. It leverages Node.js for backend operations, MongoDB for data storage, and React for the frontend. The EMS integrates InfluxDB for real-time system performance monitoring, where metrics such as CPU usage, memory consumption, and network statistics are captured, and visualised in Grafana.

Related work in this field

This system aligns with the increasing demand for real-time performance monitoring in modern web applications.

Numerous tools and studies have explored aspects of system monitoring; however, none (to our knowledge) provide the exact combination of lightweight integration, extensibility, and full-stack visibility described in this implementation.

▪ **Existing tools and approaches**
Prometheus + Grafana Stack: Prometheus is a widely adopted time-series database designed for monitoring system and service metrics. When combined with Grafana, it enables effective visualisation. However, Prometheus uses a pull-based scraping model and is often tightly integrated with container orchestration platforms like Kubernetes. In contrast, the present system uses InfluxDB with a push

model, providing greater flexibility for granular and write-intensive data collection.

ELK Stack (Elasticsearch, Logstash, Kibana): The ELK Stack is primarily used for log aggregation and visualisation. While effective in analysing API logs, it lacks the capability for real-time system-level metrics monitoring (e.g., memory, CPU, I/O) without extensive configuration.

New Relic/Datadog: These commercial observability platforms offer comprehensive features across metrics, logs, and traces. However, their closed source nature and licensing costs make them less suitable for customisable and open source deployments such as the one presented here.